MH4500 Lab 3

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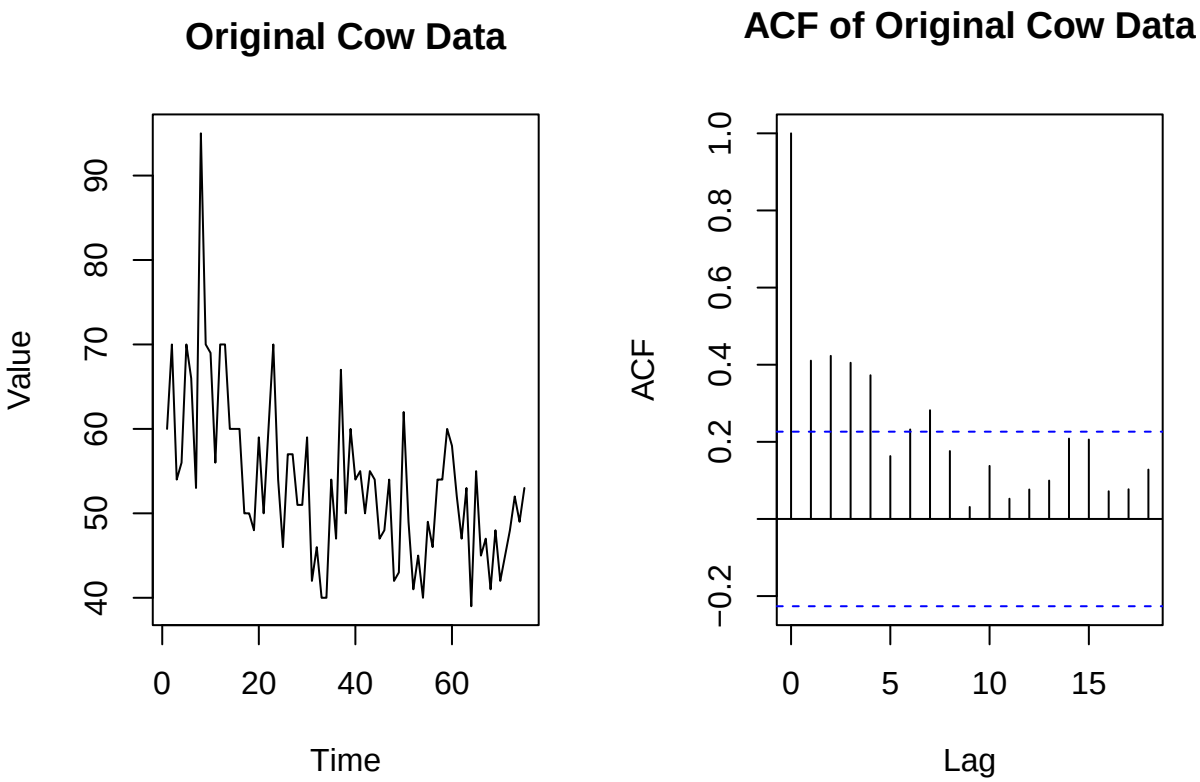
Loading packages and setting wd

```
library(forecast)

setwd('/Users/yikenchoo/Desktop/MH4500 Time Series Analysis/MH4500 Lab/MH4500 Lab 3')
```

Loading and understanding the data

```
cow <- scan("cow.dat")
par(mfrow = c(1, 2))
plot.ts(cow, main = "Original Cow Data", ylab = "Value")
acf(cow, main = "ACF of Original Cow Data")
```



According to the raw time series plot, we can see that the amplitude of fluctuations decreases over time. The fluctuations are visibly larger from time $\in$ [0,10], and decreases over time, indicating a possible non-stationary in variance present in the time series plot. The ACF plot for the raw cow data also indicates that the sample ACF slowly decays over time, indicating the raw data is not stationary. Thus, we proceed to perform Box-Cox transformation and differencing to obtain stationary data.

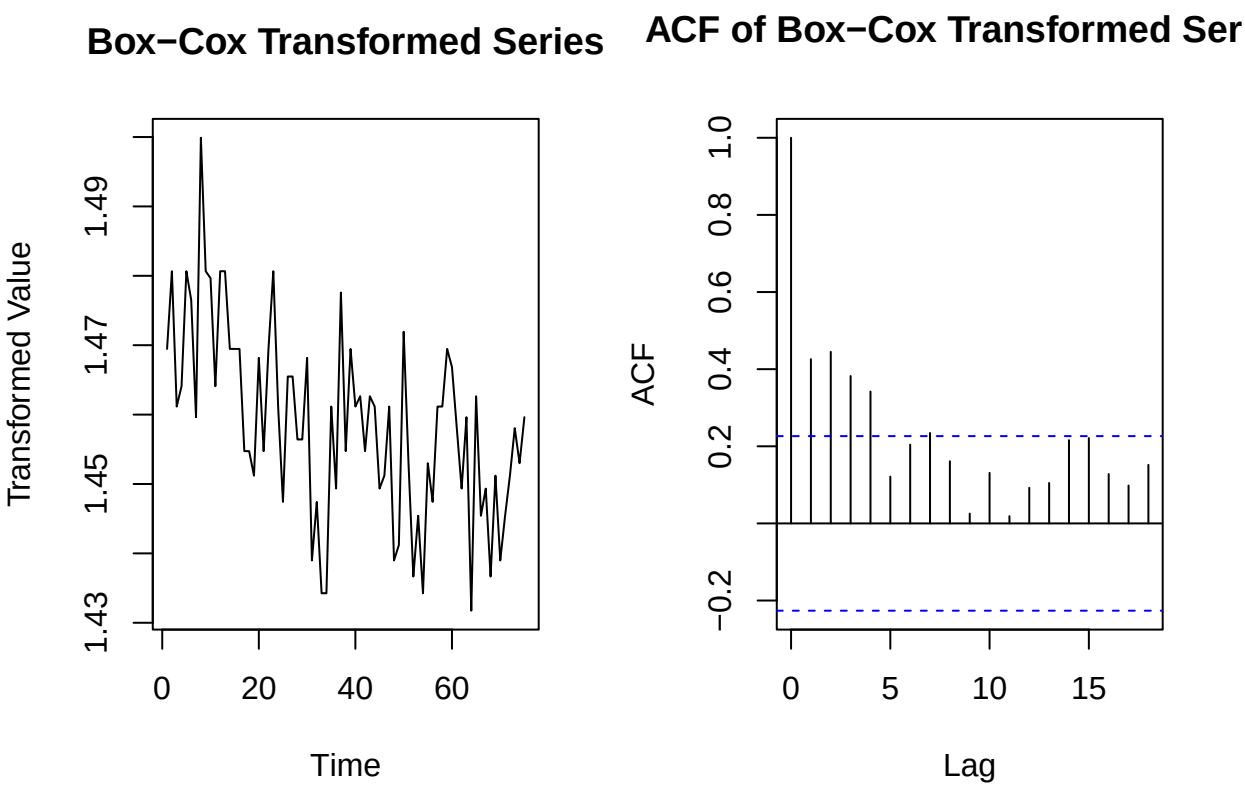
Data Transformation

Box-Cox Transformation

```
set.seed(42)
lambda <- BoxCox.lambda(cow, method = "guerrero")
cat("Optimal lambda (Guerrero method):", lambda, "\n")
```

```
## Optimal lambda (Guerrero method): -0.6286373
```

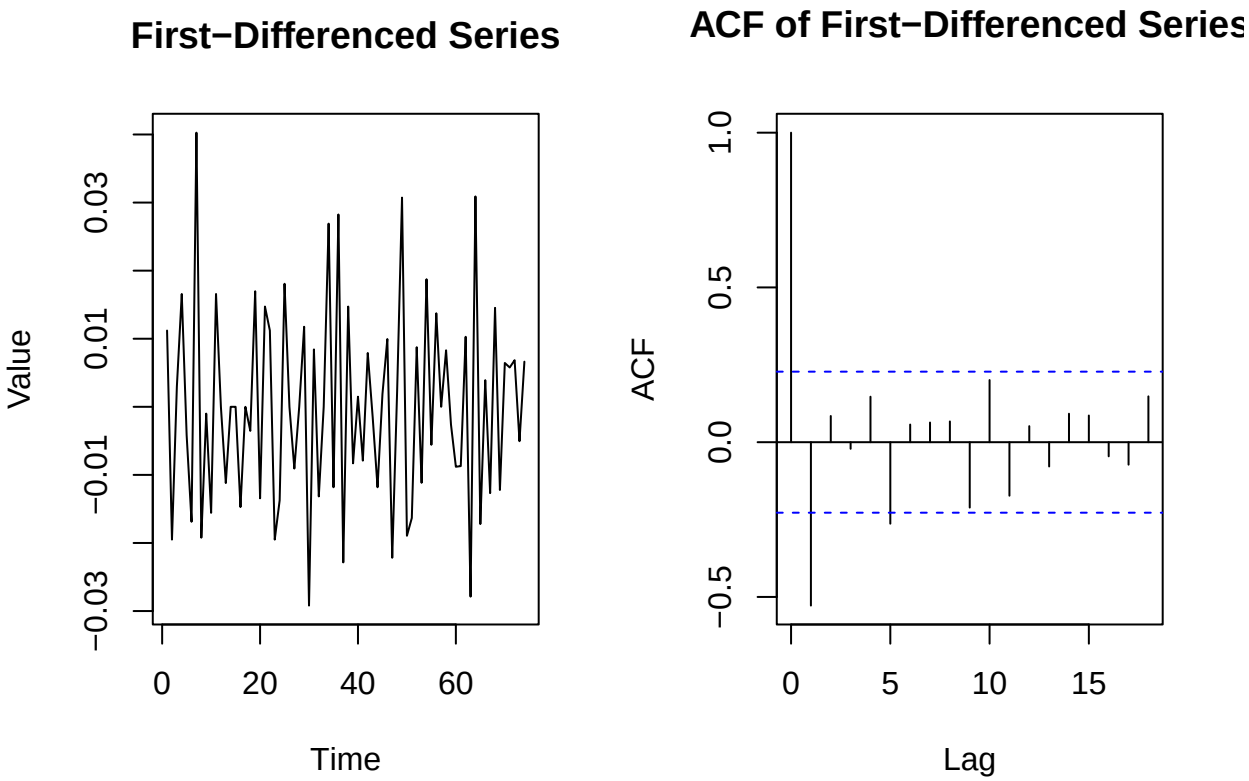
```
par(mfrow = c(1, 2))
boxcox_cow <- BoxCox(cow, lambda)
plot(boxcox_cow,
     main = "Box-Cox Transformed Series",
     xlab = "Time",
     ylab = "Transformed Value",
     type = "l")
acf(boxcox_cow, main="ACF of Box-Cox Transformed Series")
```



`BoxCox.lambda` function is loaded from the package `forecast` which returns a lambda suitable for Box-Cox transformation. Specifically, Guerrero's (1993) method is used to perform the search for lambda. While Box-Cox stabilizes the variance, the series still exhibits non-stationarity in mean, hence differencing is required.

First-Time Differencing

```
d = ndiffs(boxcox_cow)
diff_cow <- diff(boxcox_cow, differences = d)
par(mfrow = c(1, 2))
plot.ts(diff_cow, main = "First-Differenced Series", ylab = "Value")
acf(diff_cow, main = "ACF of First-Differenced Series")
```

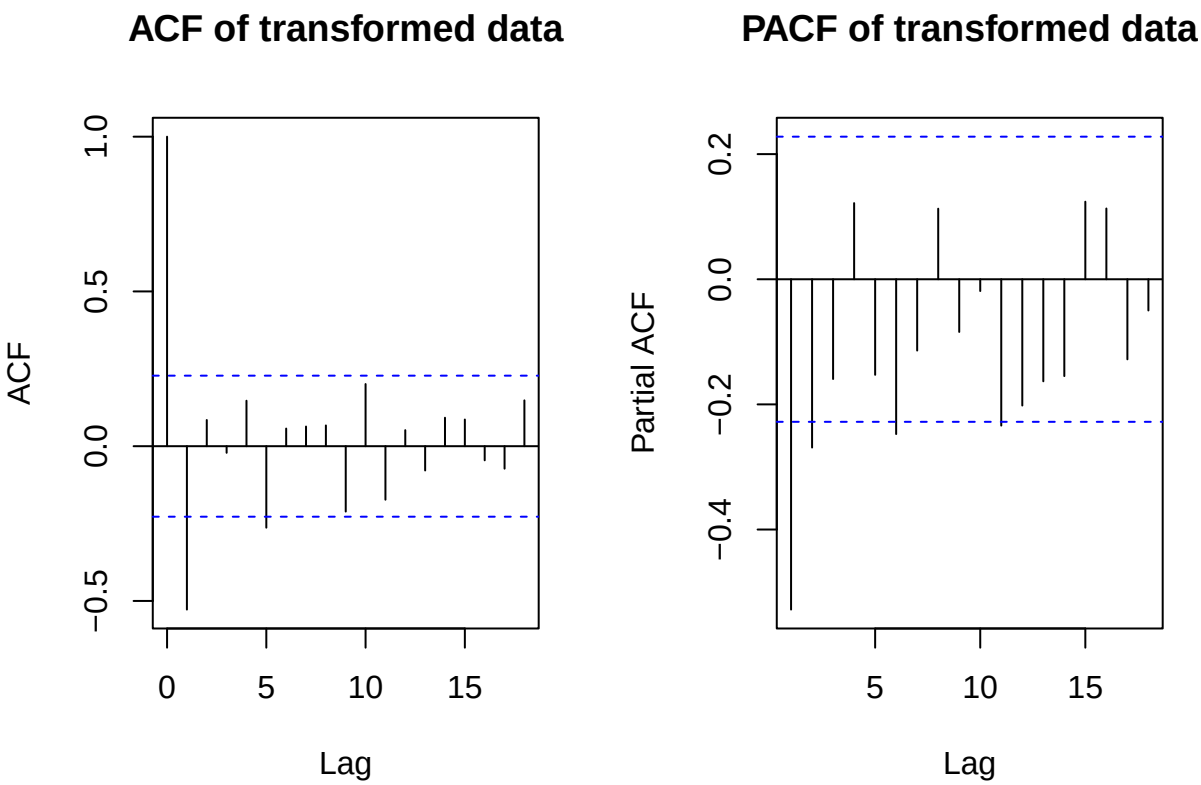


As such, we wish to apply differencing operator to the transformed data. We use `ndiffs` to estimate the number of non-seasonal differences required to make a given time series stationary. After differencing, both the time series and ACF plot shows transformed data is stationary now.

ARIMA Model Fitting

Searching for candidate models

```
par(mfrow = c(1, 2))
acf(diff_cow, main="ACF of transformed data")
pacf(diff_cow, main="PACF of transformed data")
```



Candidate models:

- ACF plot suggests MA(5) model
- PACF plot suggests AR(6) model
- We also test ARMA(6,5) model in this case

Significance Test for the Mean of the Differenced Series

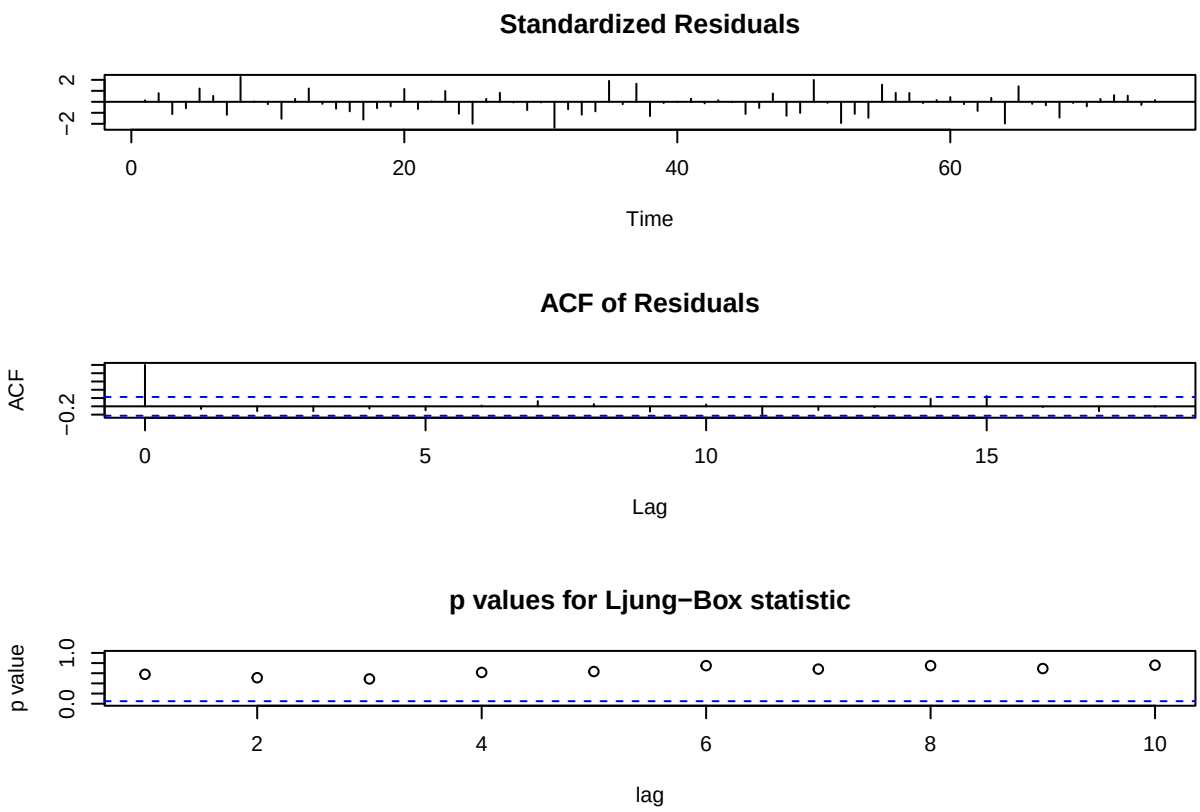
```
x_bar = mean(diff_cow)
s = sd(diff_cow)
n = length(diff_cow)

abs(x_bar / (s / sqrt(n)))
```

```
## [1] 0.07784158
```

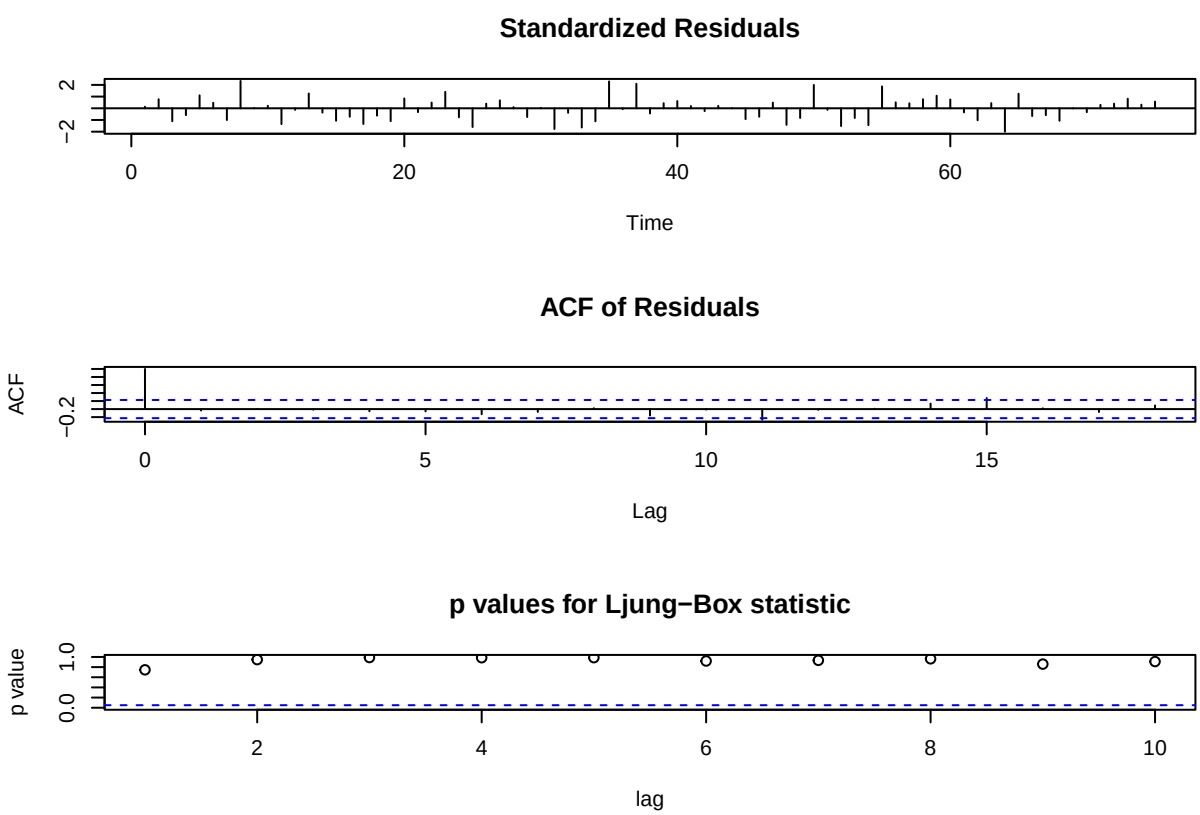
Since the test statistics ≤ 1.96, we don't reject H0 at 95% confidence level and conclude that $\delta = 0$. We will build the arima models with $\delta = 0$ in the next subsection.

Diagnostic checking

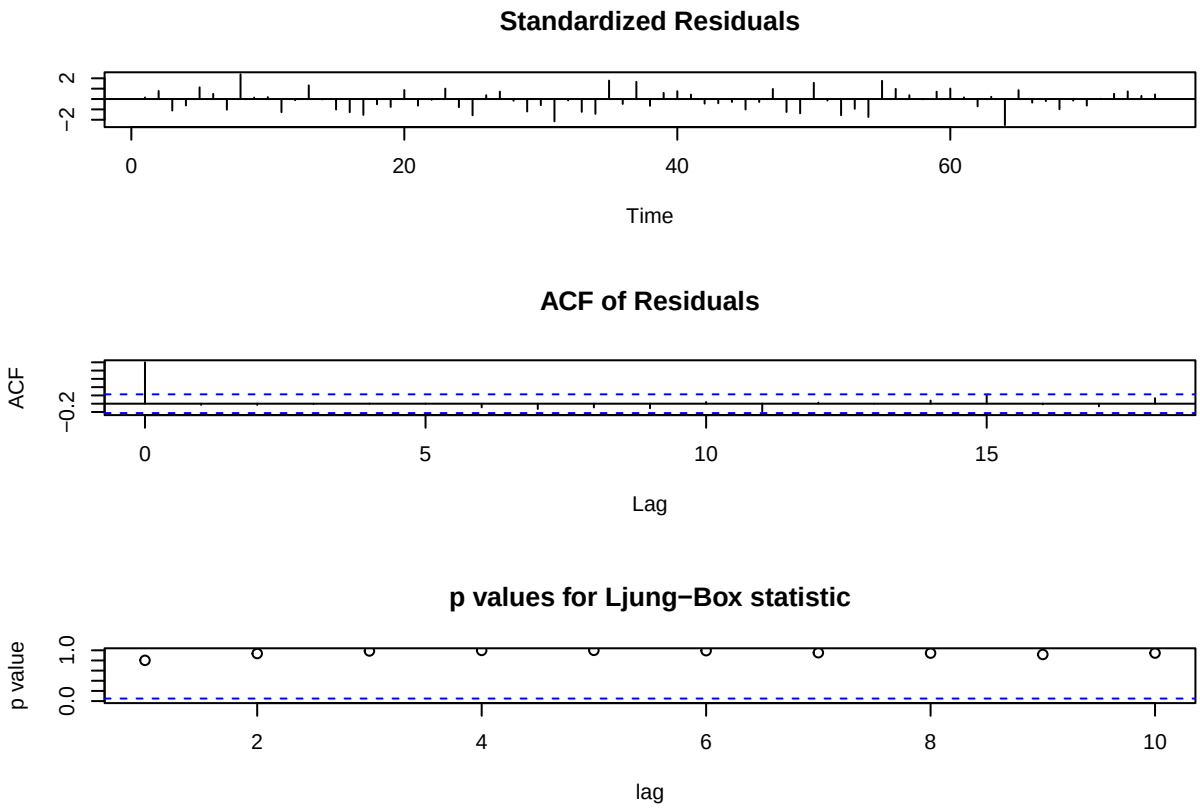
```
fit1 = arima(boxcox_cow, order = c(0,1,5), include.mean = FALSE)
fit2 = arima(boxcox_cow, order = c(6,1,0), include.mean = FALSE)
fit3 = arima(boxcox_cow, order = c(6,1,5), include.mean = FALSE)

tsdiag(fit1, main = "MA(5)")
```




```r
tsdiag(fit2, main = "AR(6)")
```



```r
tsdiag(fit3, main = "ARMA(6,5)")
```



Based on the three residuals plots, the residuals fluctuate without any significant pattern, suggesting that they are white noise. Based on Ljung-Box statistics, p value is larger than 0.05, indicating that we don't reject H0. The three models are adequate models.

Selection of models

```r
fit1
```

```
##
## Call:
## arima(x = boxcox_cow, order = c(0, 1, 5), include.mean = FALSE)
##
## Coefficients:
##           ma1     ma2     ma3      ma4      ma5
##       -0.7283  0.2535  0.0206  -0.0902  -0.3014
## s.e.   0.1105  0.1439  0.1495   0.1501   0.1712
##
## sigma^2 estimated as 0.0001203:  log likelihood = 228.04,  aic = -444.08
```

```r
fit2
```

```
##
## Call:
## arima(x = boxcox_cow, order = c(6, 1, 0), include.mean = FALSE)
##
## Coefficients:
##           ar1      ar2      ar3      ar4      ar5      ar6
##       -0.7095  -0.3373  -0.1517  -0.0732  -0.3344  -0.2573
```

```
## s.e.   0.1114   0.1330   0.1400   0.1397   0.1341   0.1122
##
## sigma^2 estimated as 0.0001234:  log likelihood = 227.41,  aic = -440.82
```

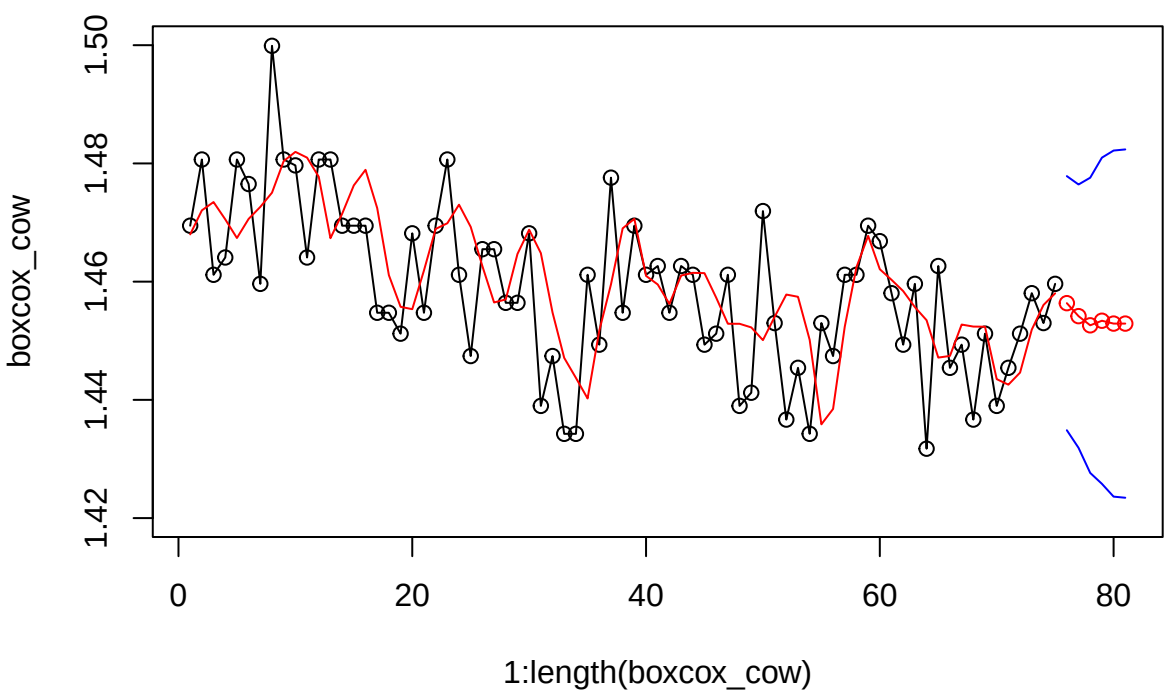
```
fit3
```

```
##
## Call:
## arima(x = boxcox_cow, order = c(6, 1, 5), include.mean = FALSE)
##
## Coefficients:
##          ar1      ar2     ar3      ar4      ar5     ar6      ma1     ma2
##       0.8558  -0.8173  0.2975  -0.2042  -0.1993  0.3011  -1.7172  1.8918
## s.e.  0.3603   0.4718  0.4718   0.3765   0.1569  0.1550   0.3729  0.6683
##           ma3     ma4      ma5
##       -1.2672  0.6443  -0.4095
## s.e.   0.9223  0.8669   0.4167
##
## sigma^2 estimated as 9.717e-05:  log likelihood = 233.73,  aic = -443.45
```

Based on the AIC values, ARIMA(0,1,5) has the lowest AIC out of the three models, thus it is selected as the most suitable model out of the three.

Plotting and Forecasting

```
plot(1:length(boxcox_cow), boxcox_cow, xlim = c(1, 81), ylim = c(1.42, 1.5))
lines(1:length(boxcox_cow), boxcox_cow)
lines(1:length(fit1$residuals), boxcox_cow-fit1$residuals, type='l', col='red')
forecast = predict(fit1, n.ahead = 6)
lines(76:81, forecast$pred, type="o", col="red")
lines(76:81, forecast$pred - 1.96*forecast$se, col="blue")
lines(76:81, forecast$pred + 1.96*forecast$se, col="blue")
```



The plot shows the Box-Cox transformed series alongside the fitted values and a 6-step ahead forecast from

the ARIMA(0,1,5) model. Black circles connected by lines are the actual observed Box-Cox transformed data over 75 data points. Red line indicates the fitted values from ARIMA(0,1,5) model.The red circles connected by lines are predicted values from t = 76 to 81, forming a 6-step ahead point forecast. Blue lines indicate the 95% confidence interval bounds for the 6 point forecasts.